To effectively cover the topics outlined in the BCA-201 Mathematics II syllabus, here are curated YouTube videos and online resources for each unit:

Unit I: Sets

- **YouTube Video:** "Discrete Mathematics #02 Sets, Relations and Functions"
    - This lecture provides an in-depth understanding of sets, subsets, universal sets, finite and infinite sets, and operations on sets.

Unit II: Relations and Functions

- **YouTube Video:** "Set Theory, Relations, Poset and Lattices"
    - This video covers properties of relations, equivalence relations, partial order relations, and functions including domain, range, and types of functions.

Unit III: Partial Order Relations and Lattices

- **Online Resource:** "Partial Orders and Lattices in Discrete Mathematics"
    - This article explains partial order sets, Hasse diagrams, chains, maximal and minimal points, lattices, and their properties.

Unit IV: Functions of Several Variables

- **Online Resource:** "Calculus III - Functions of Several Variables"
    - This tutorial discusses partial differentiation, change of variables, chain rule, extrema of functions of two variables, and Euler's theorem.

Unit V: 3D Coordinate Geometry

- **YouTube Video:** "3D Coordinate Geometry - Direction Cosines and Ratios"
    - This video explains coordinates in space, direction cosines, angles between two lines, equations of planes, straight lines, and spheres.

Unit VI: Multiple Integration

- **Online Resource:** "Calculus III - Triple Integrals"
    - This section covers double and triple integrals in Cartesian and polar coordinates, changing the order of integration, and finding volumes of simple shapes.

These resources should comprehensively cover the topics in your BCA-201 syllabus.